

1. Administrative & Study Identification

Full Study Title: A Phase II, Multicenter, Randomized, Double Masked, Active Comparator-Controlled Study to Investigate the Efficacy, Safety, Tolerability, Pharmacokinetics, and Pharmacodynamics of R07200220 Administered Intravitreally in Patients With Diabetic Macular Edema **Short Title/Acronym:** A Study to Investigate Vamikibart (R07200220) in Diabetic Macular Edema **Protocol Number:** BP43445 (EudraCT: 2021-003756-16) **NCT Number:** NCT05151731 **Sponsor Name:** Hoffmann-La Roche (Roche) **Investigational Product:** Vamikibart (R07200220) **Phase of Development:** Phase II **Trial Status:** COMPLETED **Medical Monitor:** Hoffmann-La Roche Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective **Primary Objective:** To investigate the efficacy, safety, tolerability, pharmacokinetics, and pharmacodynamics of vamikibart (R07200220) administered intravitreally compared to an active comparator (ranibizumab) in patients with diabetic macular edema (DME). **Secondary Objectives:** To evaluate the change in Central Subfield Thickness (CST) and Best-Corrected Visual Acuity (BCVA), monitor the overall reduction in diabetic retinopathy severity, and evaluate the pharmacokinetic profile and immunogenicity over the treatment period.

2.2 Background & Rationale To address the medical necessity for more durable and effective treatments for DME. Current standard-of-care therapies (like anti-VEGF agents) require frequent intravitreal injections, creating a high treatment burden. This study investigates whether vamikibart, a novel monoclonal antibody, provides meaningful clinical improvements in visual acuity and anatomical outcomes with a potentially reduced injection frequency.

3. Study Design & Methodology

3.1 Design Framework **Study Type:** Interventional **Control Type:** Active Comparator-Controlled (Ranibizumab) **Blinding:** Double-masked (Participant, Investigator/Outcomes Assessor masked) with a sham procedure administered to maintain masking across different dosing schedules. **Randomization:** Randomized (Parallel assignment)

3.2 Planned Enrollment & Duration Target Enrollment: 394 participants **Number of Sites:** 74 locations (Global and US Regional breakdown) **Estimated Study Start:** 31-Dec-2021 **Primary Completion:** 06-Nov-2024

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Diagnosis of diabetes mellitus (Type 1 or Type 2). 2. Macular thickening secondary to diabetic macular edema (DME) involving the center of the macula. 3. Decreased visual acuity attributable primarily to DME. 4. Ability and willingness to provide written informed consent and to comply with the study protocol. 5. Willingness to allow Aqueous Humor collection. 6. For women of childbearing potential: agreement to remain abstinent or use at least one highly effective contraceptive method that results in a failure rate of <1% per year during the treatment period and for at least 12 weeks after the final dose of study treatment.

4.2 Exclusion Criteria 1. Hemoglobin A1c (HbA1c) of greater than (>) 12%. 2. Uncontrolled blood pressure, defined as a systolic value greater than (>) 180 millimeters of mercury (mmHg) and/or a diastolic value > 100 mmHg while a patient is at rest. 3. Currently pregnant or breastfeeding, or intend to become pregnant during the study. 4. Prior treatment with panretinal photocoagulation or macular laser to the study eye. 5. Any intraocular or periocular corticosteroid treatment within the past 16 weeks prior to Day 1 to the study eye. 6. Prior Iluvien or Retisert implants within 3 years prior to Day 1 to the study eye. 7. Prior or concomitant treatment with anti-VEGF therapy within 8 weeks prior to Day 1 to the study eye; Vabysmo™ within 16 weeks prior to Day 1, prior Beovu® is not permitted. 8. Prior administration of IVT brolocizumab (Beovu®): ever; vamikibart: < / =24 weeks prior to Day 1 in either eye. 9. Any proliferative diabetic retinopathy. 10. Active intraocular or periocular infection or active intraocular inflammation in the study eye. 11. Any current or history of ocular disease other than DME that may confound assessment of the macula or affect central vision in the study eye. 12. Any current ocular condition which, in the opinion of the investigator, is currently causing or could be expected to contribute to irreversible vision loss due to a cause other than DME in the study eye. 13. Other protocol-specified inclusion/exclusion criteria may apply.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: * **Investigational Arm:** Vamikibart 0.25 mg or 1.0 mg intravitreal (IVT) injection on Day 1 and every 8 weeks (Q8W) or every 4 weeks (Q4W), up to Week 44 (total of 6 to 12 injections). * **Comparator Arm:** Ranibizumab (ATC Code: S01LA04; Trade names: Lucentis, Susvimo, Cimerli,

Byooviz; EMA Info: [Byooviz EPAR](#), [Lucentis EPAR](#)) 0.5 mg IVT Q4W up to Week 44 (total of 12 injections). * *Note: A sham procedure is administered during study visits at which no active drug is administered to maintain masking between treatment arms.* **Route of Administration:** Intravitreal (IVT) Injection **Duration of Treatment:** 44 weeks of treatment, followed by an observational period, bringing the total study duration up to 72-76 weeks.

5.2 Concomitant & Prohibited Therapy Permitted: Standard concomitant medications for diabetes management and glycemic control. **Prohibited:** Concurrent treatment with other ocular anti-VEGF products or intravitreal corticosteroids in the study eye.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to 0	Informed Consent, Ophthalmic Exam, Visual Acuity Assessment, Labs
Baseline	Day 0	Randomization, First IVT Dose or Sham, PK/PD Sampling
Interim	Weeks 4 to 44	IVT Injections/Sham (Q4W or Q8W), Safety Exams, Efficacy Assessment (Visual Acuity)
End of Study	Weeks 52 to 76	Observational follow-up, Final Visual Acuity & OCT, Exit Interview, Safety Labs

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Mean Change From Baseline in Best Corrected Visual Acuity (BCVA) Averaged Over Week 44 and Week 48, in Treatment-naïve Participants. **Measurement Method:** Standardized BCVA clinical visual acuity charts (e.g., ETDRS) and clinical ophthalmic evaluation.

7.2 Secondary & Exploratory Endpoints * Number of Participants With Systemic and Ocular Adverse Events (AEs). * Number of Participants With Abnormal Laboratory Findings, Abnormal Vital Signs Values, or Abnormal Electrocardiogram (ECG) Parameters. * Number of Participants With Abnormalities in Standard Ophthalmological Assessments. * Change from baseline in Central Subfield Thickness (CST) measured via Optical Coherence Tomography (OCT). * Pharmacokinetics (Cmax, AUC) and Pharmacodynamics of R07200220.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Standard collection via ophthalmic exams, including monitoring for intraocular inflammation, intraocular pressure spikes, and systemic AE events. **Serious Adverse Events (SAEs):** Reported within 24 hours to the sponsor and safety team. **Data Monitoring Committee (DMC):** Independent Data Monitoring Committee (iDMC) overseeing safety events.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) for primary efficacy outcomes. Safety Set for all AE/SAE evaluations. **Sample Size Justification:** Target enrollment of 394 participants sufficiently powered ($\alpha = 0.05$, $\text{Power} = 80\%$) to detect statistically significant differences in BCVA changes versus active comparator. **Handling of Missing Data:** Mixed-Effect Model Repeated Measure (MMRM) and Multiple Imputation (MI) methodologies for continuous efficacy endpoints.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Routine onsite and remote monitoring to ensure protocol adherence, IVT procedural safety, and valid BCVA assessments. **Audit Trail:** eCRF locking, tracking of masked reading center interpretations for imaging (OCT), and standardized data clarification forms.

11. Study Identifiers & Governance

Primary ID: BP43445 **Registry Number:** NCT05151731 **Posting ID:** 272684262554643007 **Sponsor/Lead Organization:** Hoffmann-La Roche (Roche) **Study Title:** A Study to Investigate Vamikibart (R07200220) in Diabetic Macular Edema **Official Regulatory Title:** A Phase II, Multicenter, Randomized, Double Masked, Active Comparator-Controlled Study to Investigate the Efficacy, Safety, Tolerability, Pharmacokinetics, and Pharmacodynamics of R07200220 Administered Intravitreally in Patients With Diabetic Macular Edema

12. Clinical Framework (Disease Specific Adaptation)

Primary Condition: Diabetic Macular Edema (DME) **Condition Definition:** Diabetic Macular Edema is a complication of diabetes characterized by fluid accumulation and swelling in the macula, the central portion of the retina responsible for sharp, detailed vision. It results from damaged and leaky blood vessels in the retina, caused by prolonged high blood sugar, leading to impaired central vision. **Semantic Type:** Disease or Syndrome **Terminology Codes:** MeSH Code: D000085890 | SNOMED ID: 312912001 **Diagnosis Threshold:** Macular thickening secondary to DME involving the center of the macula with decreased visual acuity **Medical Methodology:** Intravitreal (IVT) injections (Anti-IL6 vs. Anti-VEGF Standard of Care) **Optimization Target:** Best-Corrected Visual Acuity (BCVA) improvement and macular thickness (CST) reduction

13. Study Procedures & Methodology

Management Pathway: Standard clinical pathway for DME investigational treatment. **Education Protocol (HCP):** Masked sham procedure protocol training and IVT injection safety protocols. **Education Protocol (Patient):** Trial regimen compliance, understanding of sham vs. active injections, and reporting ocular symptoms. **Quality Audit Mechanism:** Masked central reading centers for OCT interpretations and certified masked assessors for visual acuity (ETDRS) evaluation.

14. Observation & Follow-Up Schedule

Total Duration: Up to 76 weeks **Onsite Visit Cadence:** Every 4 weeks (Q4W) for dosing/sham injections and assessments. **Remote Monitoring Frequency:** Standard AE follow-up as required. **Checklist Review Interval:** Ophthalmic safety checklist and systemic AE review conducted at every Q4W/Q8W onsite dosing visit.

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]		Lead		
Statistician	[Name]	_____	[DD-MMM-YYYY]				